CLINICAL · LARYNX & AIRWAY

Tracheostomy and airway endoscopy

The tracheostomy is one of the oldest operations in surgery and still one of the most important: it bypasses an obstructed upper airway, shortens the dead space, allows the chest to be cleared and ventilators to be weaned. Done calmly on an intubated patient it is a safe, orderly procedure; done in a hurry on a struggling, hypoxic patient it is one of the most dangerous in the specialty. Know the anatomy, know the alternatives that buy time, and know how to look after the tube, because most tracheostomy deaths happen afterwards, on the ward, from a blocked or displaced tube.

AT A GLANCE

- Never through the first ring.
- Do not change the tube before a track forms.
- Cricothyroidotomy for can't intubate, can't oxygenate.
- Rigid bronchoscopy for foreign bodies.
- NTSP bedside algorithms and capnography; tracheo-innominate fistula needs cuff overinflation and theatre.

1 CHAPTER ONE Tracheostomy

Purpose and indications.

1.1 Definitions

Tracheotomy is opening the trachea; *tracheostomy* is making that opening into a stoma at the skin. The terms are used interchangeably.

1.2 What a tracheostomy does

1. **Bypasses obstruction** anywhere from the lips to the stoma.
2. **Improves alveolar ventilation** by reducing dead space by 30 to 50% (normal dead space about 150 mL) and lowering airway resistance.
3. **Protects the airway** with a cuffed tube against aspiration of secretions (coma, bulbar palsy) and blood (maxillofacial and laryngeal trauma), and allows the pharynx

to be packed.

4. **Allows suction** of secretions when the patient cannot cough (coma, head injury, respiratory paralysis) or coughing hurts (chest and upper abdominal injury or surgery).
5. **Permits prolonged ventilation:** for intermittent positive pressure ventilation expected beyond a week or two (Dhingra suggests beyond 72 hours), tracheostomy is better tolerated than translaryngeal intubation and protects the larynx.
6. **Allows anaesthesia** when intubation is impossible (trismus, obstructing tumours).

1.3 Indications

INDICATIONS FOR TRACHEOSTOMY

GROUP	EXAMPLES
Upper airway obstruction: infection	Epiglottitis, croup, diphtheria, Ludwig's angina, quinsy, retropharyngeal and parapharyngeal abscesses
Obstruction: trauma	Laryngotracheal injury, maxillofacial and mandibular fractures, endoscopic and intubation injury
Obstruction: other	Tumours of the larynx, pharynx, tongue and thyroid; foreign bodies; laryngeal oedema (burns, irritants, allergy, radiation); bilateral abductor paralysis; congenital lesions (webs, cysts, subglottic stenosis, haemangioma)
Retained secretions	Coma, head injury, stroke, overdose; respiratory muscle paralysis (spinal injury, Guillain–Barré, myasthenia, polio); tetanus; painful cough; aspiration
Respiratory insufficiency	Prolonged ventilation; chronic lung disease (emphysema, bronchitis, bronchiectasis)
Prophylactic	Major surgery of the tongue, floor of mouth, mandible, pharynx and larynx; obstructive sleep apnoea unresponsive to other treatment

In children under a year most indications are congenital (subglottic stenosis, haemangioma, cysts, webs, bilateral paralysis); in older children inflammatory or traumatic (croup, epiglottitis, diphtheria, burns, prolonged intubation, papillomatosis).

2 CHAPTER TWO Types

When and where.

2.1 By urgency

- **Emergency:** obstruction is complete or nearly so and intubation is impossible; ideally replaced by cricothyroidotomy or by intubation followed by an orderly tracheostomy.
- **Elective (orderly):** planned, with the patient intubated where possible; therapeutic or prophylactic.
- **Permanent:** an end stoma after laryngectomy, or a long-term tube for bilateral paralysis or stenosis.
- **Percutaneous dilatational:** at the bedside in intensive care.
- **Minitracheostomy and cricothyroidotomy.**

2.2 By level

The thyroid isthmus lies over the second to fourth tracheal rings.

- **High** (above the isthmus, through the first ring): risks perichondritis of the cricoid and subglottic stenosis, and is avoided; it is acceptable only in laryngeal cancer, when the stoma will be removed at laryngectomy.
- **Mid** (through the second to fourth rings, dividing or retracting the isthmus): the preferred site.
- **Low** (below the isthmus): the trachea is deep, close to great vessels, and the tube presses on the suprasternal notch.

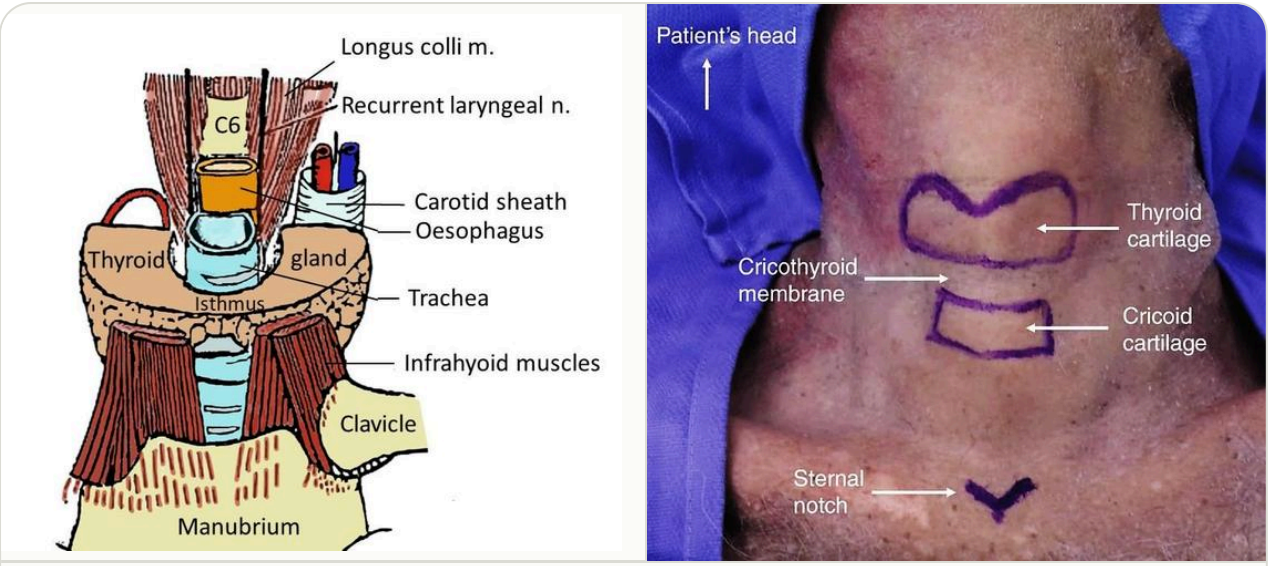


FIGURE 1 The cervical trachea and recurrent laryngeal nerve, and surface marking of the thyroid and cricoid cartilages and cricothyroid membrane.

Anatomy atlas · Doss p. 176, ENT Steps p. 262

3 CHAPTER THREE Technique

The open operation.

3.1 Preparation

Intubate first whenever possible, especially in children. Supine with a shoulder roll to extend the neck and bring the trachea forward (unless the cervical spine is injured). Local infiltration with lidocaine and adrenaline in the awake patient; general anaesthesia when intubated.

3.2 Steps

1. **Incision:** a vertical midline incision from the cricoid to just above the sternal notch gives fast access with little bleeding and suits emergencies; a 4 to 5 cm horizontal incision two fingerbreadths above the notch gives a better scar in elective cases.
2. Dissect in the midline; retract or tie the anterior jugular veins.
3. Separate the strap muscles in the midline and retract them laterally.
4. Retract the thyroid isthmus, or divide it between clamps and suture-ligate it.

5. Instil a little lidocaine into the trachea to suppress coughing; steady the trachea with a cricoid hook.
6. Open the trachea at the second to fourth rings: a vertical slit, a window, an inverted U (Björk) flap sutured to the skin to ease reinsertion, or a slit with stay sutures; never through the first ring.
7. Withdraw the endotracheal tube to just above the opening under direct vision, insert the tracheostomy tube, inflate the cuff, confirm position (capnography), and only then remove the endotracheal tube.
8. Secure the tube with tapes (and sutures); do not close the skin tightly or pack the wound, which causes surgical emphysema.

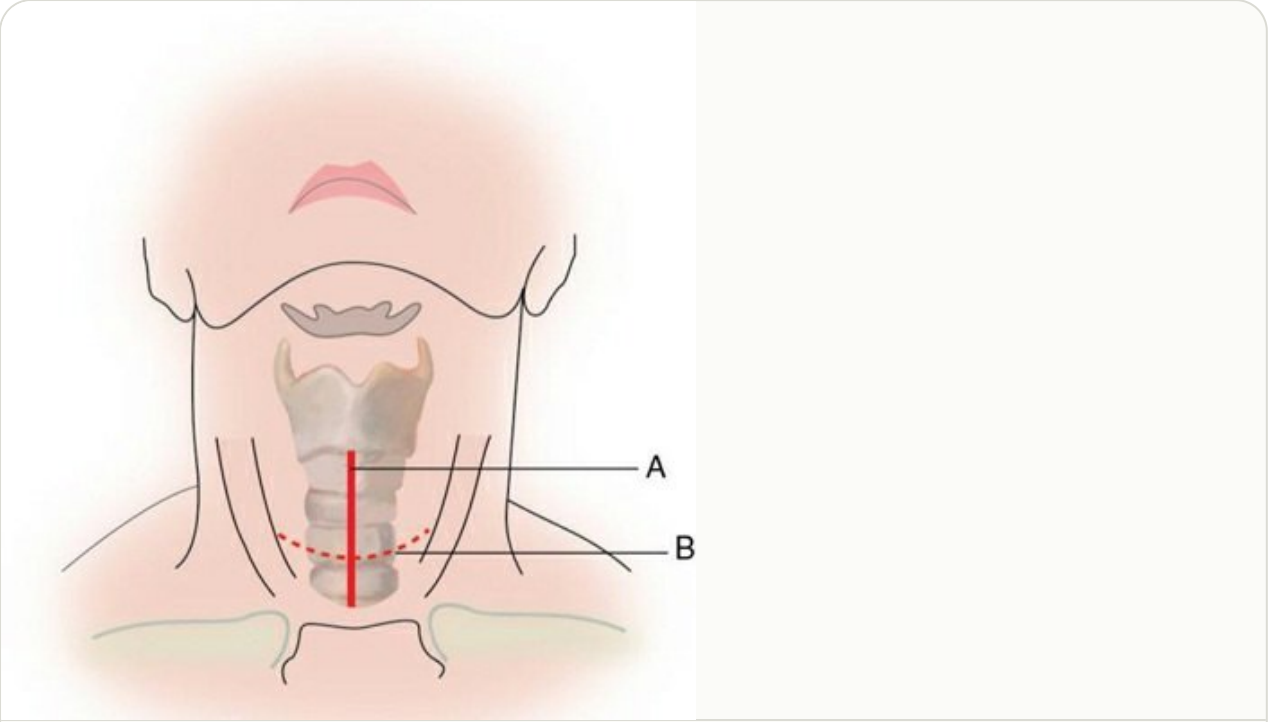


FIGURE 2 Skin incisions for tracheostomy: (A) vertical midline, (B) transverse.
Dhingra, fig. 64.1

3.3 In infants and children

- Operate under general anaesthesia over an endotracheal tube or bronchoscope; the trachea is soft and easily lost, and straying laterally injures the recurrent nerve and carotid.
- Do not hyperextend the neck: it draws the pleura, innominate vessels and thymus into the field and invites too low an opening.
- Place *stay sutures* on either side of the midline before incising; they are the rescue if the tube comes out in the first days.

- Make a simple vertical incision; never remove cartilage.
- Do not push the blade deep: the posterior wall and oesophagus lie close.
- Choose a soft tube of the right diameter, length and curvature; confirm its position with a chest radiograph.

3.4 Percutaneous dilatational tracheostomy

In the intubated intensive care patient under sedation, through a small incision, a needle, guide wire and dilators (single tapered or serial) enter between the second and third rings under bronchoscopic guidance through the endotracheal tube. It avoids transfer to theatre and is quicker and cheaper, with fewer wound infections. Avoid in obesity, neck masses, difficult intubation, poor extension, impalpable landmarks, coagulopathy and in children. Complications: paratracheal placement, bleeding, posterior wall injury and emphysema.

OPERATIVE GUIDES IN THEATRE 2 operations

Open and percutaneous tracheostomy.

Open Tracheostomy
Open tracheostomy

Percutaneous Tracheostomy
Percutaneous tracheostomy

4 CHAPTER FOUR Tracheostomy tubes

Choosing the tube.

FEATURES OF TRACHEOSTOMY TUBES

FEATURE	PURPOSE
Cuffed	Seals the airway for ventilation and against aspiration; keep pressure below about 25 to 30 cm of water to avoid mucosal necrosis
Uncuffed	For patients who no longer need ventilation or protection; children
Inner cannula	Removed and cleaned to prevent blockage without changing the tube

FEATURE	PURPOSE
Fenestrated	A window allows air to pass up through the larynx for speech when the inner cannula is removed and the tube occluded
Speaking (one-way) valve	Lets air in through the tube and out through the larynx (only with the cuff deflated)
Adjustable flange	For deep or obese necks
Metal (silver, Jackson)	Durable, for long-term permanent use; no cuff



FIGURE 3 A patient with a suction-aid tracheostomy tube receiving oxygen through a side port. Dhingra, fig. 64.3

5 CHAPTER FIVE Aftercare

Where safety is won or lost.

5.1 The first days

1. **Constant observation** by trained staff: bleeding, displacement, blockage and secretions; a call bell and writing pad.
2. **Suction** as often as needed with a sterile catheter, applying suction only while withdrawing it, to avoid mucosal injury.
3. **Humidification** of inspired air (heated humidifiers, nebulisers, heat-moisture exchangers) to prevent crusting and tracheitis; saline instillation and mucolytics for

crusts.

4. **Inner cannula** cleaned regularly; the *outer tube is not changed for the first three to five days* until a track forms, after which tube changes become straightforward.
5. **Cuff** pressure checked; deflated when no longer needed.
6. Spare tubes (same size and smaller), tracheal dilator, obturator, suction and oxygen at the bedside.

THE BLOCKED OR DISPLACED TUBE

Sudden respiratory distress in a patient with a tracheostomy: remove the speaking valve and inner cannula, pass a suction catheter; if it will not pass, deflate the cuff and remove the tube; oxygenate through the stoma and the face; if the patient was not a laryngectomy, they can also be oxygenated and intubated from above. Follow a standard emergency algorithm and colour-coded bedhead signs.

5.2 Decannulation

Remove the tube as soon as it is not needed, because prolonged use brings infection, ulceration, granulation, stenosis and scarring. Deflate the cuff, change to a smaller uncuffed tube, then occlude (cap) it; if the patient breathes, speaks and clears secretions comfortably for 24 hours (longer in children), remove the tube and tape the stoma, which usually closes within a week; persistent tracheocutaneous fistulae are closed surgically. In infants, decannulate in hospital with full reintubation equipment and monitoring.

5.3 Difficult decannulation

Look for persistence of the original disease, granulations at the stoma or the tube tip (often a suprastomal granuloma), subglottic stenosis or oedema, collapse of the anterior wall above the stoma, tracheomalacia, and psychological dependence; endoscopy of the larynx and trachea decides.

6 CHAPTER SIX Complications

Immediate, early and late.

6.1 At operation

- Haemorrhage (anterior jugular veins, isthmus, aberrant vessels).
- *Apnoea* after relief of long-standing obstruction, when the high carbon dioxide drive is suddenly removed; ventilate.
- Pneumothorax and pneumomediastinum (especially in children).
- Recurrent laryngeal nerve injury; injury to the oesophagus with a tracheo-oesophageal fistula.
- Aspiration of blood; false passage; cardiac arrest.

6.2 Early (hours to days)

- Reactionary and secondary bleeding.
- *Displacement* and *blockage* of the tube: the commonest causes of death.
- Surgical emphysema, tracheitis and crusting, atelectasis and pneumonia, wound infection.

6.3 Late

- *Tracheo-innominate fistula* with massive haemorrhage from erosion by the tube tip or cuff, often heralded by a sentinel bleed.
- Subglottic and laryngeal stenosis (high tracheostomy), tracheal stenosis at the stoma or cuff site, tracheomalacia.
- Tracheo-oesophageal fistula from cuff pressure or the tube tip.
- Granulations, difficult decannulation, persistent tracheocutaneous fistula, ugly or keloid scar.
- Corrosion and fracture of old metal tubes with aspiration of fragments.

OPERATIVE GUIDES IN THEATRE 2 operations

Tracheocutaneous fistula and depressed scar, and stenosis of the tracheostoma after laryngectomy.

Management of a Tracheocutaneous Fistula and Depressed Scar
Tracheocutaneous fistula

Stenosis of the Tracheostoma Following Total Laryngectomy
Stomal stenosis

7 CHAPTER SEVEN Emergency airway

Buying time.

When obstruction leaves no time for an orderly tracheostomy:

1. **Jaw thrust** and head tilt (not in spinal injury) lift the tongue off the pharyngeal wall.
2. **Oropharyngeal airway** with bag-and-mask ventilation.
3. **Nasopharyngeal airway:** better tolerated by an awake patient.
4. **Supraglottic airway (laryngeal mask):** a rescue when mask ventilation and intubation fail; it does not help obstruction at or below the glottis.
5. **Intubation:** the quickest definitive airway; then an orderly tracheostomy if needed.
6. **Cricothyroidotomy:** through the cricothyroid membrane, between the thyroid and cricoid cartilages, in the "can't intubate, can't oxygenate" situation. *Scalpel cricothyroidotomy* (a stab through the membrane, rotated, with a bougie and a 6 mm cuffed tube) is the recommended adult technique. Convert to a formal tracheostomy within about 72 hours, because perichondritis and subglottic stenosis follow prolonged cricothyroidotomy. It is avoided in children under about ten to twelve, whose membrane is tiny.
7. **Needle cricothyroidotomy** with a large cannula and *transtracheal jet ventilation*: a temporary measure, the choice in small children; ensure a path for expiration or barotrauma follows.
8. **Emergency tracheostomy:** with the trachea fixed between thumb and finger, a vertical incision from the thyroid to the notch, quickly through to the second and third rings; hazardous on a struggling patient without light, suction or help, so intubate first if at all possible.



FIGURE 4 A laryngeal mask airway.
Dhingra, fig. 64.4

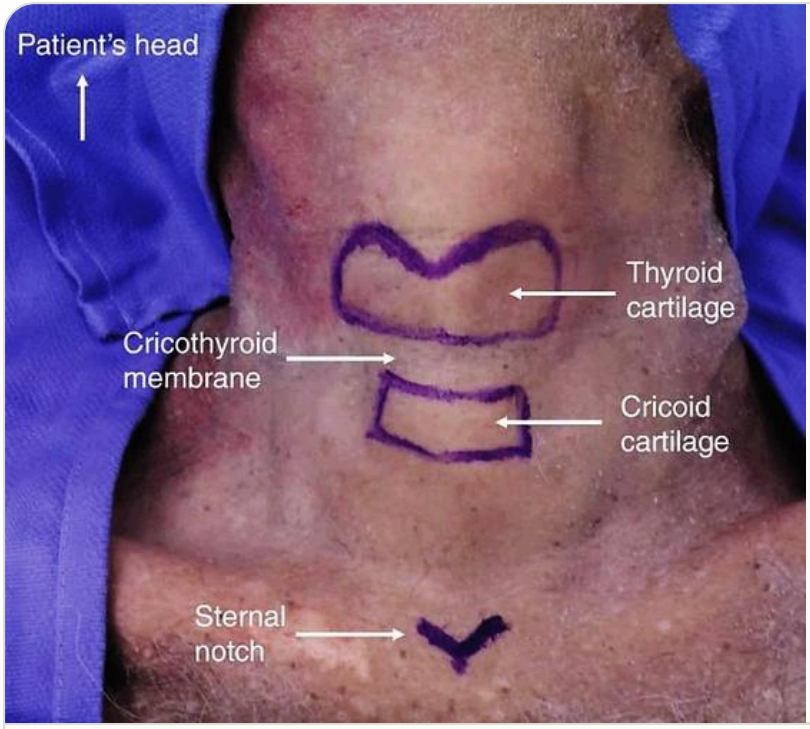


FIGURE 5 Surface marking of the cricothyroid membrane.
Anatomy atlas · ENT Steps p. 262

8 CHAPTER EIGHT Airway endoscopy

Laryngoscopy and bronchoscopy.

8.1 Direct laryngoscopy

To see the larynx and hypopharynx directly: when indirect examination is impossible (children, a strong gag, an overhanging epiglottis), to examine the *hidden areas* (base of tongue, valleculae, lower piriform sinuses, infrahyoid epiglottis, anterior

commissure, ventricles and subglottis), to assess and biopsy tumours, and to treat (remove benign lesions and foreign bodies, dilate strictures). Contraindicated by cervical spine disease and by significant obstruction unless the airway has first been secured.

- **Position:** the *Boyce (sniffing, barking-dog) position*: head raised 10 to 15 cm on a pillow with the neck flexed on the chest and the head extended at the atlanto-occipital joint, aligning the oral, pharyngeal and laryngeal axes.
- **Technique:** protect the upper teeth; insert the laryngoscope along the right side of the tongue, move to the midline, lift the epiglottis, and examine every subsite systematically, lifting rather than levering on the teeth. Suspension and the operating microscope or angled telescopes allow *microlaryngoscopy*.
- **Complications:** injury to lips, teeth and tongue, bleeding, laryngeal oedema, laryngospasm.

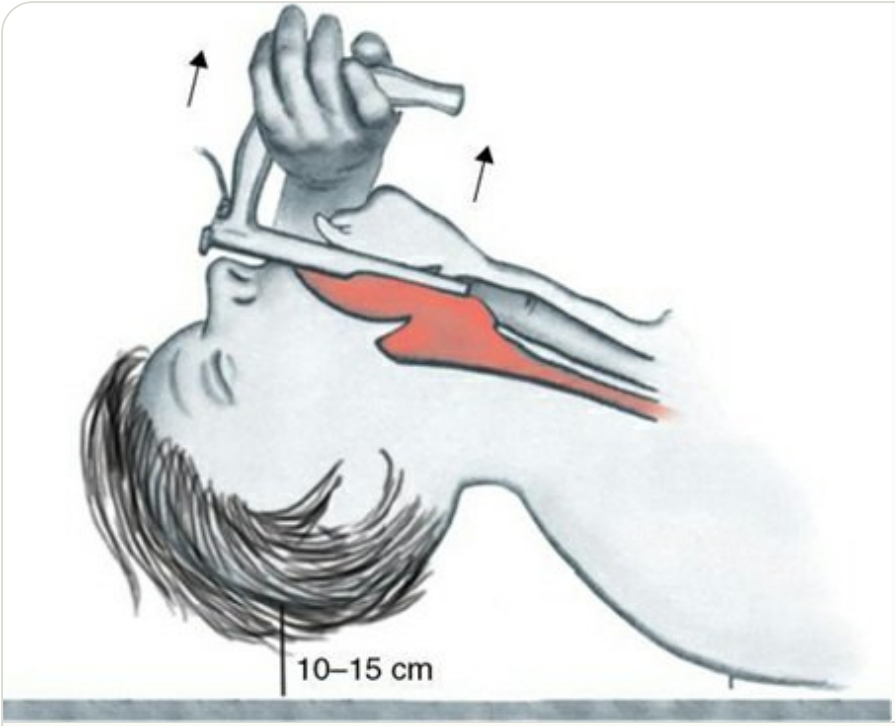


FIGURE 6 Direct laryngoscopy in the sniffing position, lifting the laryngoscope without levering on the teeth.
Dhingra, fig. 91.1

8.2 Bronchoscopy

- **Indications:** unexplained wheeze, haemoptysis or cough over four weeks; radiographic collapse, persistent opacity, obstructive emphysema, hilar or mediastinal

shadows; vocal fold palsy; secretions for culture and cytology. Therapeutic: foreign bodies, retained secretions and mucus plugs.

- **Rigid bronchoscopy** under general anaesthesia allows ventilation through the scope, large biopsies, control of haemorrhage and is the instrument of choice for *foreign bodies*, especially in children. The bevel is turned to pass the glottis, and the head is moved to the left to enter the right bronchus and vice versa. Limit procedures in children to about twenty minutes to avoid subglottic oedema.
- **Flexible bronchoscopy** under topical anaesthesia (through the nose, an endotracheal tube or tracheostomy) reaches segmental and subsegmental bronchi, can be done at the bedside and in patients with neck or jaw problems, but controls ventilation and bleeding poorly and is not the tool for foreign bodies in children.

OPERATIVE GUIDES IN THEATRE 3 operations

Bronchoscopy, tracheoscopy and airway foreign bodies, and oesophagoscopy.

Bronchoscopy
Bronchoscopy
Tracheoscopy, Bronchoscopy, and Airway Foreign Bodies
Esophagoscopy and Hypopharyngoscopy

SAFETY AND QUALITY (OXFORD TEXTBOOK)

Tracheostomy problems account for up to half of airway-related hypoxic brain injuries and deaths in hospital, and most are preventable. The UK **National Tracheostomy Safety Project** emergency algorithms for a blocked or displaced tube (separate versions for patients with a patent upper airway and for laryngectomy "neck breathers") should be at every bedside, with bedside equipment, a sign showing the airway type, and staff trained to use them; the **Global Tracheostomy Collaborative** promotes the same team-based approach internationally. Waveform **capnography** confirms correct placement at surgery and after any tube change. In the first days, before a tract forms, displacement is an emergency: stay sutures and a Björk flap make reinsertion easier, and if the tube cannot be replaced the patient is oxygenated from above and intubated orally.

Preserving the thyroid isthmus where possible shortens surgery without adding complications. In a patient likely to need a laryngectomy, a **high** tracheostomy is preferred, so that the stoma can be excised with the specimen; the theoretical risk of tumour seeding is not clinically important. A thyroid ima artery is present in up to 5% and a high-riding innominate artery can be injured low in the neck. Tracheal stenosis is promoted by cuff pressures above 30 cmH_2O, intubation beyond ten days, reflux, obesity, large tubes and percutaneous insertion. A **tracheo-innominate fistula** occurs in under 1% but kills over 80%; a sentinel bleed demands immediate

assessment, and torrential bleeding is controlled by overinflating the cuff or by digital pressure through the stoma against the sternum while the patient goes to theatre.

9 CHAPTER NINE For the examination

What the examiner wants to hear.

9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Functions and indications** of tracheostomy.
- **Why not through the first ring;** the levels.
- **Steps** of the operation, and differences in children.
- **Tubes,** aftercare, decannulation and causes of failed decannulation.
- **Complications,** especially displacement, blockage and tracheo-innominate fistula.
- **Emergency airway options:** cricothyroidotomy and jet ventilation.
- **Direct laryngoscopy** position and hidden areas; **rigid versus flexible bronchoscopy.**

9.2 Common pitfalls

COMMON PITFALLS

- Making a high tracheostomy through the first ring.
- Changing the tube on day one without a mature track or stay sutures.
- Suturing the skin tightly around the tube.
- Leaving a cricothyroidotomy in place for weeks.

Written for this app from Oxford Textbook of Otolaryngology ch. 70, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 20, Dhingra ch. 64, 91, 92 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · TRAUMA

Laryngeal Trauma

IMAGING FINDINGS

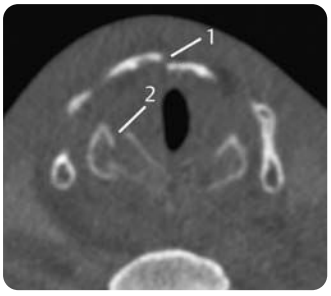
- Thyroid and cricoid cartilage fractures with displacement
- Arytenoid dislocation
- Haematoma and air narrowing the airway

TOP DIFFERENTIALS

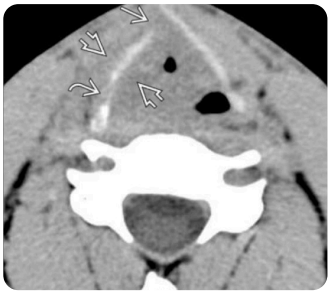
- Cord paralysis
- Unossified cartilage mimicking a fracture

CHECKLIST

- Secure airway before scanning
- Cricoid fractures and displaced fragments usually need repair



CT, axial bone window. CT at the level of the glottis showed multiple fractures; this figure shows a midline fracture with slight inward…



Axial NECT through the supraglottic larynx shows step-off at the superior thyroid notch. There is a subtle fracture of the right thyroid lamina…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 278.

PHARYNX, TONSILS & OESOPHAGUS · TRAUMA

Foreign Bodies

IMAGING FINDINGS

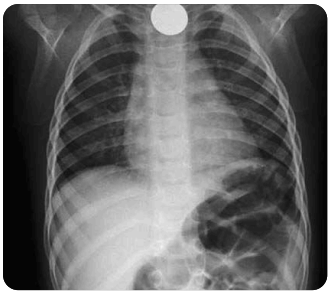
- Radio-opaque body on the lateral neck film; fish bones often faint
- CT shows the object, its position and depth
- Prevertebral swelling, air or abscess after perforation

TOP DIFFERENTIALS

- Calcified cartilage (thyroid, cricoid, arytenoid)
- Calcified stylohyoid ligament

CHECKLIST

- Low-dose CT if the film is negative and suspicion is high
- Perforation: mediastinal air, collection



Conventional radiography can still be convenient for evaluation of radiopaque foreign bodies, as in this 2-year-old child, who was suspected of…